Isaac Padberg

Personal Website LinkedIn

(802) 585-7241 - iaverypadberg@gmail.com

Professional Experience

Software Engineer at Deep Analytics - Jan 2022 - Aug 2026(4.5yrs)

Machine Learning & Computer Vision

- Trained and evaluated models including YOLO, SSD, GANs, and Stable Diffusion for object detection and generation tasks.
- Managed and optimized datasets. Designed sanitation and processing pipelines for specific use cases, implementing them with FiftyOne and CVAT.
- Architected model inference pipelines for embedded devices.
- Improved model performance using synthetic data, leveraging Segment Anything Model (SAM) for efficient dataset generation.
- Worked with Docker, PyTorch, TensorFlow, Keras, OpenCV, and NumPy to develop and deploy models.

Software & Deployment

- Led the development of a $2M website upgrade to integrate real-time object detection models.
- Designed and implemented real-time feedback UIs, enabling dynamic inference parameter customization for improved model performance and user control.
- Developed and maintained containerized applications for model deployment and scaling.

Additional Contributions

- Assisted in proposal writing for R&D initiatives and funding opportunities.
- Strong self-starter, capable of independently solving complex technical challenges.

Projects

Bad Road(For Cycling) Identification

- Created a website which uses historical traffic data and user supplied road reviews to create a catalog of roads that should be avoided. Anyone can upload a GPX file which will cross check your route with the bad road database and flag dangerous roads.

Exploratory Analysis of behavior and emotions

- Conducted a self study to investigate how behavior influences emotions
- Collected 6 months of personal data and used a correlation matrix to search for patterns.
- Wrote a 1-pager review of my work and outlined how I intend to build on this existing project.

Technical Skills & Formal Education

Specialties: Computer Vision, Website Development

ML Languages & Libraries: Python (Ultralytics, Pytorch, SciKitLearn, Pandas, Tensorflow, Numpy, Keras,Fiftyone), Kotlin(Tensorflow & Pytorch)

WebDev Languages & Libraries: Javascript(Reactjs,Vuejs), HTML, CSS(Tailwind), PHP(Laravel), SQL, AWS

Technologies/Platforms: Git(Gitlab/Github), Docker, Figma, Nuclio, Android Studios, VSCode

Education: B.A. in CS, University of Vermont, 2017-2021.